Group II, Row IV: extraction of the BFKL, JIMWLK and finite (DGLAP + RC) terms

1 Check of the starting expression

The starting expression is

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^++\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'} e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]\frac{1}{\xi^2}$$
$$\left[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\,\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)$$
$$+\;\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^++\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'} e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\xi^2}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\,\xi(1-\xi)\right.$$
$$\left.+\;\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\left[\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]\right]\,\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\,. \qquad (1)$$

It is correct: it coincides exactly with the reduction of the Row-IV master expression (colour algebra $U^{d'b}(z)U^{db}(z)=\delta^{d'd}$, $f^{c'da}f^{cda}=N_c\delta^{c'c}$; δ-function integration over w,w'; contraction of the two tensor brackets over i,j, in which the antisymmetric $\frac{2}{k^+}(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km})$ channel cancels; the dimensionally regularized z-integral $\int_z \mathcal{K}^m(y-z)\mathcal{K}^{m'}(y'-z)=-\frac{\pi}{2}L_\epsilon\delta^{mm'}-\pi\frac{r^m r^{m'}}{r^2}$, $r=y'-y$, $L_\epsilon\equiv\frac{2}{\epsilon}+2\gamma_E+\log\frac{r^2\mu^2_{\overline{MS}}}{4}$; and the substitution $\xi=\frac{k^+}{p^++k^+}$), as derived step by step in the companion note *"Group II, Row IV: reduction of the master expression"*. Two features were checked explicitly: (i) the ξ-limits are written *descending*, from $\xi_{\max}=\frac{k^+}{k^++\Lambda}$ down to $\bar\xi=\frac{k^+}{\vee}$, which absorbs the sign of $dp^+=-\frac{k^+d\xi}{\xi^2}$, so all the printed signs are right; (ii) the placement of $d_\perp$ is right: it multiplies the full $\mathcal{P}$-combination in the L_ϵ term, but only the azimuthal kernel in the second term — the dipole piece $\left[\frac{1-\xi}{\xi}+\frac{\xi}{1-\xi}\right]$ carries no $d_\perp$, since it originates from the $-\pi\,\delta_{mm'}\frac{r^m r^{m'}}{r^2}=-\pi$ trace of the z-integral.

2 Notation and first regrouping

Write $C\equiv\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}=\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}$, $f(\xi)=e^{-ik(y'-y)/\xi}$, $\hat p(\xi)=\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{1-\xi}$, and keep the colour bracket of (1) implicit. Flipping the limits to ascending ($\int_{\xi_{\max}}^{\bar\xi}=-\int_{\bar\xi}^{\xi_{\max}}$), Eq. (1) reads

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\,C\int_{\bar\xi}^{\xi_{\max}}d\xi\,\Big\{\frac{d_\perp}{2}\,\frac{\hat p(\xi)}{\xi^2}\,L_\epsilon+\frac{1}{\xi^2}\Big[\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big]\Big\}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,f(\xi)-C\,d_\perp\int_{\bar\xi}^{\xi_{\max}}d\xi\,\frac{1-\xi}{\xi}\,\frac{[\mathcal{K}(x-y)\cdot r][\mathcal{K}(x'-$$
(2)

Using the exact identity

$$\frac{\hat p(\xi)}{\xi^2}-\frac{1-\xi}{\xi}=\frac{1}{\xi^2}\Big[\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big]\,, \qquad (3)$$

the L_ϵ-free dipole piece is rewritten as $\hat p/\xi^2$ minus $\frac{1-\xi}{\xi}$, giving

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\,C\int_{\bar\xi}^{\xi_{\max}}d\xi\,\frac{\hat p(\xi)}{\xi^2}\,\underbrace{\Big[\frac{d_\perp}{2}L_\epsilon+1\Big]}_{=\,L_\epsilon\,+\,O(\epsilon)}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')f(\xi)\;+\;C\int_{\bar\xi}^{1}d\xi\,\frac{1-\xi}{\xi}\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')-d_\perp\,\frac{(\mathcal{K}\cdot r)(\mathcal{K}'\cdot r)}{r^2}\Big]f(\xi)$$
(4)

where $\frac{d_\perp}{2}L_\epsilon+1=\frac{d_\perp}{\epsilon}+d_\perp\gamma_E+\frac{d_\perp}{2}\log\frac{r^2\mu^2}{4}+1=L_\epsilon$ for $d_\perp=2-\epsilon$ (the $+1$ from $-\frac{\epsilon}{2}\cdot\frac{2}{\epsilon}$ cancels the explicit $+1$), and the upper limit of the regular second integral has been extended to 1 at a cost $O(\Lambda)$.

3 Endpoint extraction: the JIMWLK term

A $\log\Lambda$ can only come from the upper limit $\xi_{\max}=\frac{k^+}{k^++\Lambda}\simeq 1-\frac{\Lambda}{k^+}$. Of the three pieces of $\hat{p}(\xi)$, $\xi(1-\xi)$ and $\frac{1-\xi}{\xi}$ are regular at $\xi=1$, so for them $\xi_{\max}\to 1$ at a cost $O(\Lambda)$. The entire Λ-dependence of (4) therefore sits in the single channel $\frac{\xi}{1-\xi}$, which is singular at $\xi\to 1$ — and $\xi\to 1$ means $p^+\to 0$: *the unmeasured gluon becomes soft.* The logarithm regulated by Λ at this endpoint is the target-side rapidity logarithm, i.e. the JIMWLK piece. Isolating that channel and writing its integrand as $h(\xi)/(1-\xi)$ with

$$h(\xi)\equiv\Big[\frac{d_\perp}{2}L_\epsilon+1\Big]\,\frac{f(\xi)}{\xi}\,, \tag{5}$$

we add and subtract the endpoint value $h(1)$ in the numerator:

$$\int_{\bar\xi}^{\xi_{\max}}d\xi\,\frac{h(\xi)}{1-\xi}=\int_{\bar\xi}^{\xi_{\max}}d\xi\,\frac{h(\xi)-h(1)}{1-\xi}\;+\;h(1)\int_{\bar\xi}^{\xi_{\max}}\frac{d\xi}{1-\xi}\,. \tag{6}$$

In the first piece the numerator vanishes linearly at $\xi=1$, so the integrand is finite there and $\xi_{\max}\to 1$ at a cost $O(\Lambda)$: this finite remainder is what the locally subtracted $\frac{\xi}{(1-\xi)_+}$ denotes. The second piece is elementary:

$$\int_{\bar\xi}^{\xi_{\max}}\frac{d\xi}{1-\xi}=\log\frac{1-\bar\xi}{1-\xi_{\max}}=\log\frac{\big(1-\frac{k^+}{\vee}\big)(k^++\Lambda)}{\Lambda}=\log\frac{(\vee-k^+)\,k^+}{\vee\,\Lambda}+O(\Lambda)\,, \tag{7}$$

Sign check of (7). The antiderivative of $\frac{1}{1-\xi}$ is $-\log(1-\xi)$, so substituting the limits in the standard order, *upper minus lower*,

$$\int_{\bar\xi}^{\xi_{\max}}\frac{d\xi}{1-\xi}=\Big[-\log(1-\xi)\Big]_{\xi=\bar\xi}^{\xi=\xi_{\max}}=-\log(1-\xi_{\max})-\big[-\log(1-\bar\xi)\big]=\log\frac{1-\bar\xi}{1-\xi_{\max}}\;:$$

it is the minus sign of the antiderivative that puts the *lower* limit in the numerator of the ratio; the limits themselves are substituted the usual way around. Consistency check: the integrand $\frac{1}{1-\xi}$ is positive on the whole range and the limits are ascending ($\bar\xi<\xi_{\max}$), so the integral must be positive — and indeed $1-\bar\xi=\frac{\vee-k^+}{\vee}=O(1)$ while $1-\xi_{\max}=\frac{\Lambda}{k^++\Lambda}\to 0$, so (7) is large and positive. (The descending limits of the original expression (1) were already traded for the overall minus sign of (2); no second sign flip occurs here.)

The endpoint value $h(1)$. The only ξ-dependence left in h is the explicit $\frac{1}{\xi}$ and the phase $f(\xi)=e^{-ik(y'-y)/\xi}$: the transverse kernels $\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')$ and the bracket $\frac{d_\perp}{2}L_\epsilon+1$ carry no ξ at all. The $\frac{1}{\xi}$ in the phase is purely kinematic: in the master expression the phase is $e^{-ik(y'-y)\frac{p^++k^+}{k^+}}$, and $\frac{p^++k^+}{k^+}=\frac{1}{\xi}$. At the endpoint $\xi=1$, i.e. $p^+=0$, the unmeasured gluon carries zero longitudinal momentum, so the measured gluon retains the full leading-order kinematics: the phase collapses to the LO phase, $f(1)=e^{-ik(y'-y)}$, and the explicit $\frac{1}{\xi}\to 1$. Hence

$$h(1)=\Big[\frac{d_\perp}{2}L_\epsilon+1\Big]\,e^{-ik(y'-y)}\,,\qquad\log\frac{(\vee-k^+)k^+}{\vee\Lambda}=\log\frac{k^+}{\Lambda}+\log\Big(1-\frac{k^+}{\vee}\Big)\,. \tag{8}$$

The coefficient of the endpoint logarithm is thus the L_ϵ-dressed LO dipole — the soft unmeasured gluon is the JIMWLK gluon. This channel enters (4) with the prefactor $-C$, so its endpoint piece is $-C\,h(1)\times(7)$; pulling the minus sign into the bracket, $-h(1)=\big[-\frac{d_\perp}{2}L_\epsilon-1\big]e^{-ik(y'-y)}$, which is where the overall + sign and the all-negative bracket of the box below both come from (one minus sign, used once — it originates from flipping the descending limits of (1), not from the limit substitution in (7)). The $\log\frac{k^+}{\Lambda}$ part is the **JIMWLK contribution of this row**,

$$\frac{d^3N^{\text{JIMWLK}}_{\text{G-II, R IV}}}{d^3k}=+\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\,\log\Big(\frac{k^+}{\Lambda}\Big)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\Big[-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}-1 \tag{9}$$

(the bracket is $-L_\epsilon$), which is moved to the two-ρ evolution (JIMWLK) file, where it appears verbatim. The $\log(1-\frac{k^+}{\vee})$ part of (8) stays here.

4 The standard plus prescription: DGLAP and RC terms

With the *standard* plus distribution on $[0,1]$,

$$\int_a^1 d\xi\,\frac{h(\xi)}{(1-\xi)_+}=\int_a^1 d\xi\,\frac{h(\xi)-h(1)}{1-\xi}+h(1)\,\log(1-a)\,,\qquad a=\bar\xi=\frac{k^+}{\vee}\,, \tag{10}$$

the locally subtracted integral of (8) and the leftover $h(1)\log(1-\frac{k^+}{\vee})$ term recombine into the single integral $\int_{\bar\xi}^1 d\xi\, h(\xi)/(1-\xi)_+$: the two groupings differ only by where the finite piece $h(1)\log(1-\bar\xi)$ is placed. This conversion changes nothing, but it is required for

$$\mathcal{P}_{gg,+}(\xi) = 2N_c\left[\xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{(1-\xi)_+}\right], \qquad \mathcal{P}_{gg}(\xi) = \mathcal{P}_{gg,+}(\xi) + \frac{\beta_0}{2}\,\delta(1-\xi)\,, \qquad \beta_0 = \frac{11N_c}{3}\ \text{(pure glue)} \tag{11}$$

to be the *standard* splitting function. Using $C = \frac{g^2}{8\pi^2}\cdot 2N_c\cdot\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}$ and defining

$$\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\,, \tag{12}$$

the first term of (4) minus the JIMWLK term becomes $-\frac{g^2}{8\pi^2}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg,+}(\xi)\,L_\epsilon\,\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}$. Finally, adding and subtracting the $\frac{\beta_0}{2}\delta(1-\xi)$ piece,

$$-\frac{g^2}{8\pi^2}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg,+}\,L_\epsilon\,\frac{d^3N_{\xi\mathrm{LO}}}{d^3k} = \underbrace{-\frac{g^2}{8\pi^2}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,L_\epsilon\,\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}}_{\text{DGLAP}} + \underbrace{\frac{g^2\beta_0}{16\pi^2}\,L_\epsilon\,\frac{d^3N_{\mathrm{LO}}}{d^3k}}_{\text{RC}}\,, \tag{13}$$

which makes the running-coupling (RC) term explicit; the two added pieces cancel identically, so this is an exact add-and-subtract.

5 The soft subtraction: the BFKL term

Step 1: where the second rapidity logarithm hides. The DGLAP term (13) contains the soft channel $2N_c\frac{1-\xi}{\xi}$ of $\mathcal{P}_{gg}$, whose lower endpoint is $\bar\xi = \frac{k^+}{\vee}$. Now $\xi\to 0$ means $p^+\to\vee$: it is the *unmeasured* gluon that carries almost all of the projectile longitudinal momentum, and the *measured* gluon is the soft one — the mirror image of the JIMWLK endpoint of Sec. 3. This is the projectile-side rapidity region, so the associated logarithm is $\log\frac{\vee}{k^+}$ and must be BFKL. In position space the $\xi\to 0$ region is the collapse $y = \xi w + (1-\xi)z \to z$, $y' = \xi w' + (1-\xi)z\to z$ of the parent positions onto the sibling at fixed w, w'. The danger is that in (13) this channel multiplies L_ϵ: left as it stands, it would generate $\log\frac{\vee}{k^+}\times\big(\frac{2}{\epsilon}+\log\mu^2+\dots\big)$, and a rapidity logarithm with an ultraviolet coefficient is not physical — the BFKL kernel is finite in $d=2$ and μ-independent. The cure is to subtract from the soft channel an expression with the *same* $\xi\to 0$ limit that can be computed exactly, and to add its value back.

Step 2: the subtraction term. The $\xi\to 0$ limit is subtracted *exactly* with

$$\frac{d^3N_\xi^{\mathrm{sub}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\pi}\,K(y,y',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\,, \tag{14}$$

$K(a,b,z) = \frac{(a-z)\cdot(b-z)}{(a-z)^2(b-z)^2}$. This is the real-emission structure with the parent sitting at the common point z: the kernels $K(y,y',z)\,K(x,x',z)$ are the two soft-emission (dipole) kernels of amplitude and conjugate, and the Wilson-line factor is taken at coincident points, $W(z,z)$ — exactly the building blocks of the real BFKL kernel. It is IR and UV finite in $d=2$, contains no μ, and is symmetric under $(x,y)\leftrightarrow(x',y')$.

Step 3: the combination is free of $\frac{2}{\epsilon}$ and μ. Writing $W(u,u') = [U^{ce'}(u')-U^{ce'}(x')][U^{ce}(u)-U^{ce}(x)]\rho^{e'}(x')\rho^{e}(x)$ and using the dimensionally regularized identity $\int_z^{(d=2-\epsilon)} K(y,y',z) = -\pi L_\epsilon$ to put the L_ϵ of the soft channel itself under a z-integral, the per-ξ combination

$$2N_c\,\frac{1-\xi}{\xi}\Big[L_\epsilon\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k} + \frac{d^3N_\xi^{\mathrm{sub}}}{d^3k}\Big] = -\frac{2N_c(1-\xi)}{\xi}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int e^{-ik(y'-y)\,\frac{1}{\xi}}\,\frac{K(y,y',z)}{\pi}\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y') - K(x,z$$

puts both pieces under one z-integral. To see what this buys, change variables at fixed z,

$$y = z + \xi\,(w-z)\,, \qquad y' = z + \xi\,(w'-z)\,, \tag{16}$$

which is an exact substitution, not an approximation. Under it

$$d^2y\,d^2y' = \xi^4\,d^2w\,d^2w'\,, \qquad K(y,y',z) = \frac{K(w,w',z)}{\xi^2}\,, \qquad e^{-ik(y'-y)\,\frac{1}{\xi}} = e^{-ik(w'-w)}\ \ \textit{exactly}, \text{ since } y'-y = \xi\,(w'-w)\,. \tag{17}$$

At fixed (w, w', z) the square bracket of (15) behaves as

$$\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y') - K(x,x',z)\,W(z,z) \xrightarrow[\xi\to 0]{} \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,W(z,z) - K(x,x',z)\,W(z,z) = 0\,, \tag{18}$$

because $y, y' \to z$ and $\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) = K(x,x',z)$ identically: the bracket vanishes *linearly* in ξ (the kernels and the Wilson lines are smooth at z). Moreover, in the variables (w, w', z) the z-integral of (15) converges at large z — the parent points y, y' travel with z, so the two terms of the bracket fall off together — which is why the combination can be evaluated directly in $d = 2$: it contains no $\frac{2}{\epsilon}$, no γ_E and no μ. And because the bracket is $O(\xi)$, the soft region contributes $\int_{\bar\xi} \frac{d\xi}{\xi} \times O(\xi) =$ finite: no $\log \frac{\vee}{k^+}$ survives in the combination. All the ultraviolet data and the entire rapidity logarithm of the soft channel are carried by the subtraction term alone, which we now evaluate.

Step 4: the exact value of the subtraction. Eq. (14) depends on ξ only through the phase, so the same substitution (16) computes it in closed form: the Jacobian ξ^4 times the kernel rescaling ξ^{-2} gives $\frac{d^3N_\xi^{\rm sub}}{d^3k} = \xi^2 \times$ (ξ-independent integral), and the ξ^2 cancels the explicit $\frac{1}{\xi^2}$ of the DGLAP measure, leaving only

$$\int_{\bar\xi}^1 d\xi\,\frac{1-\xi}{\xi} = \Big[\log\xi - \xi\Big]_{\xi=\bar\xi}^{\xi=1} = -\log\bar\xi - (1-\bar\xi) = \log\frac{\vee}{k^+} - \Big(1 - \frac{k^+}{\vee}\Big) \tag{19}$$

(upper minus lower again; the result is positive, as it must be for a positive integrand). Hence

$$-\frac{g^2}{8\pi^2}\int_{\bar\xi}^1 \frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,\frac{d^3N_\xi^{\rm sub}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}\Big[\log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,W(z,z)\,. \tag{20}$$

Step 5: add and subtract. The subtraction term was *added* to the cross-section as the "soft subtraction" entry of the finite block (23) (that entry is verbatim the left-hand side of (20); it is what renders the soft channel of the DGLAP term safe via (15)), so its exact value (20) must be *subtracted* back to change nothing. Since (20) carries an overall minus sign, subtracting it back means *adding* $+\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\big[\log\frac{\vee}{k^+} - \big(1-\frac{k^+}{\vee}\big)\big]\int K\,K\,W(z,z)$: the $\log\frac{\vee}{k^+}$ part (overall $+$) is the **BFKL contribution of this row**,

$$\frac{d^3N^{\rm BFKL}_{\text{G-II, R IV}}}{d^3k} = +\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z) - U^{ce'}(x')\Big]\,[U^{ce}(z) - U^{ce}(x) \tag{21}$$

moved to the BFKL file (it appears there verbatim); its coefficient is manifestly UV finite and μ-independent, as a BFKL kernel must be. The $-\big(1-\frac{k^+}{\vee}\big)$ part remains as the boundary term below.

6 Result: the finite terms

Collecting everything, Eq. (1) decomposes as

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{d^3N^{\rm BFKL}}{d^3k}\Big|_{(21)} + \frac{d^3N^{\rm JIMWLK}}{d^3k}\Big|_{(9)} + \frac{d^3N^{\rm finite}}{d^3k}\,, \tag{22}$$

with the finite part containing the DGLAP and RC terms:

$$\frac{d^3N^{\rm finite}}{d^3k} = \underbrace{-\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2}{8\pi^2}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,L_\epsilon\,\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-}_{\text{DGLAP}}$$

$$\underbrace{-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\frac{(1-\xi)}{\xi}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,K(y,y',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x}_{\text{soft subtraction}}$$

$$\underbrace{+\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2\,\beta_0}{16\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,L_\epsilon\,\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)}_{\text{RC}}$$

$$\underbrace{-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\Big(1-\frac{k^+}{\vee}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x'}_{\text{boundary}}$$

$$\underbrace{+\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\right]}_{\text{azimuthal}} \tag{23}$$

with $\mathcal{P}_{gg}$ in the *standard* $[0,1]$ plus-prescription convention (10)–(11), $L_\epsilon = \frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}$, and $d_\perp \to 2$ set in the azimuthal term (no pole multiplies it).

Bookkeeping

(i) (soft subtraction) + (boundary) + (BFKL entry (21)) = 0 identically — the construction is an exact add-and-subtract of (20); all three objects are UV finite. The physical origin of the BFKL logarithm is the $\xi \to 0$ endpoint of the DGLAP term.

(ii) The $\frac{\beta_0}{2}\delta(1-\xi)$ piece inside $\mathcal{P}_{gg}$ cancels the RC term identically; the soft $2N_c\frac{1-\xi}{\xi}$ piece of the DGLAP term combines with the subtraction into the μ-free form (15). The only surviving ultraviolet logarithms are $L_\epsilon \otimes 2N_c\big[\frac{\xi}{(1-\xi)_+}+\xi(1-\xi)\big]$ — the physical collinear (DGLAP) kernel of the measured gluon, including its $(1-\xi)_+$ endpoint $\log(1-\frac{k^+}{\vee})$ piece — and the RC term; these enter the final pole accounting (coupling renormalization with $\beta_0 = \frac{11N_c}{3}$ together with the Group-I ultraviolet terms, plus the collinear subtraction for the produced gluon).

(iii) Together with (9) and (21), Eq. (23) reproduces the starting expression (1) identically, up to $O(\Lambda)$ (endpoint recombination and extension of regular ξ-integrals to $\xi = 1$) and $O(\epsilon) \times$ finite.